CSMFAB000000000113-Vol-20

Thermal Management: ZrB2-SiC Spear in Hypersonic Flow

Series: CSMFAB000000000113 | **Volume:** 20 of 29 | **June 2026**

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1. Aerodynamic Heating Fundamentals

1.1 Stagnation Point Temperature

For a body moving at Mach M through atmosphere at temperature T_static:

```
T_stag = T_static * (1 + 0.5 * (gamma-1) * M^2)  [gamma=1.4 for air]
       = T_static * (1 + 0.2 * M^2)
```

Mach	Altitude (km)	T_static (K)	T_stag (K)	T_stag (°C)	ZrB2-SiC OK?
1	0	288	346	73	YES
3	10	223	624	351	YES
5	15	217	1302	1029	YES
8	20	217	3317	3044	MARGINAL
10	25	222	5106	4833	NO (melt at 3245°C)
12	30	227	7259	6986	NO

ZrB2-SiC is adequate to Mach 8-9 in the lower atmosphere. For Mach 10+ flight: nose cone and fin leading edges require HfB2.

1.2 Heat Transfer to Surface

Fay-Riddell stagnation point heat flux:

```
q_s = C_FR * (rho_e * mu_e)^0.5 * (h_aw - h_w) / sqrt(R_nose)
```

Simplified (Detra-Hidalgo-Pallone correlation):

```
q_s = 1.1e6 * (rho/rho_SL)^0.5 * (v/v_orbit)^3.15 / sqrt(R_nose)  [W/m2]
```

At M=5, h=15 km, v=1500 m/s, R_nose=0.01m:

```
q_s = 1.1e6 * (0.195/1.225)^0.5 * (1500/7800)^3.15 / 0.1
    = 1.1e6 * 0.399 * 0.00413 / 0.1 = 18,150 W/m2 = 18.15 kW/m2
```

1.3 Radiation Cooling

The ZrB2-SiC surface can radiate heat:

```
q_rad = epsilon * sigma * T_wall^4
```

For T_wall = 1500 K, epsilon = 0.85 (ZrB2 emissivity):

```
q_rad = 0.85 * 5.67e-8 * 1500^4 = 0.85 * 5.67e-8 * 5.06e12 = 243,900 W/m2
```

The radiation cooling capacity (243 kW/m2) far exceeds the aero-heating rate (18 kW/m2) up to Mach 5!

At Mach 8 where q_s 1.5 MW/m2: Required T_wall for radiation balance: T_wall = (q_s / (epsilon * sigma))^0.25 = (1.5e6 / 4.82e-8)^0.25 = 2460 K

ZrB2-SiC is stable at 2460 K (2187°C) — well below its 3245°C melt point. **ZrB2-SiC is self-cooling by radiation through Mach 10 in sparse upper atmosphere.**

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2. HfB2 Nose Cone Insert for Scale 4-5

When spear enters dense lower atmosphere at Mach 10+ (Scale 4-5 with second stage):

Hafnium Diboride (HfB2) properties: - Melting point: 3387°C (highest of any diboride) - Density: 11.2 g/cm3 - Oxidation resistance: excellent (forms HfO2 protective layer) - Thermal conductivity: 11 W/m · K

Nose cone geometry: - Ogive nose cone, half-angle 15° - HfB2 occupies front 30 mm (critical stagnation zone) - ZrB2-SiC constitutes remaining 90mm (to body join) - Mass penalty: pi*(0.04)^2*0.03*11200/2 = 0.85 kg (for 8cm radius nose)

Ablation protection: At max heating (3600 K surface temperature at Mach 10): HfO2 oxidation layer growth: 0.02 mm/s → forms passive protective layer Thermal ablation rate (net loss): 0.015 mm/s (less than HfO2 regrowth) For 10 s at peak heating: total ablation < 0.15 mm — **structurally negligible**

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3. Spear Thermal State Through Full Trajectory

Thermal history for Scale 3 (Small, v_exit=1380 m/s):

Event	Time (s)	Temperature (°C)	Status
Pre-launch (underwater)	0	10	Ambient ocean
Water-air transition	0.01	10	No heating
Mach 1 (340 m/s, h=0)	0.2	50	Slight warming
Mach 3 (900 m/s, h=5km)	2.1	320	ZrB2 comfortable
Mach 4 (h=15km)	8.5	620	ZrB2 comfortable
Apogee (100km)	85	-20 (radiates to space)	ZrB2 comfortable
Second stage burn (300°C)	95-123	400	Comfortable
Orbital insertion	150	-50 (deep space)	Stable

No thermal management issues for Scale 1-3. Scale 4-5 requires HfB2 nose insert for Mach 8+ protection.

Citations (Vol 20)

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